

Project Progress Presentation-II
on

Automated Mobile Robotic Arm for Cold Storage Operations



Presented By

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Project Guide

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Suggestions from Previous Presentation

- Suggestion:** Show calculations for the components and the system.
- Compliance:** We calculated torque, power consumption, battery capacity, and load capacity to ensure the proper functioning of the robotic arm in a cold storage environment.

Progress – Components Acquired

- **Purchased:**

- 3 MG995 servo motors

- Servo controller

- Battery

- Micro servo SG90

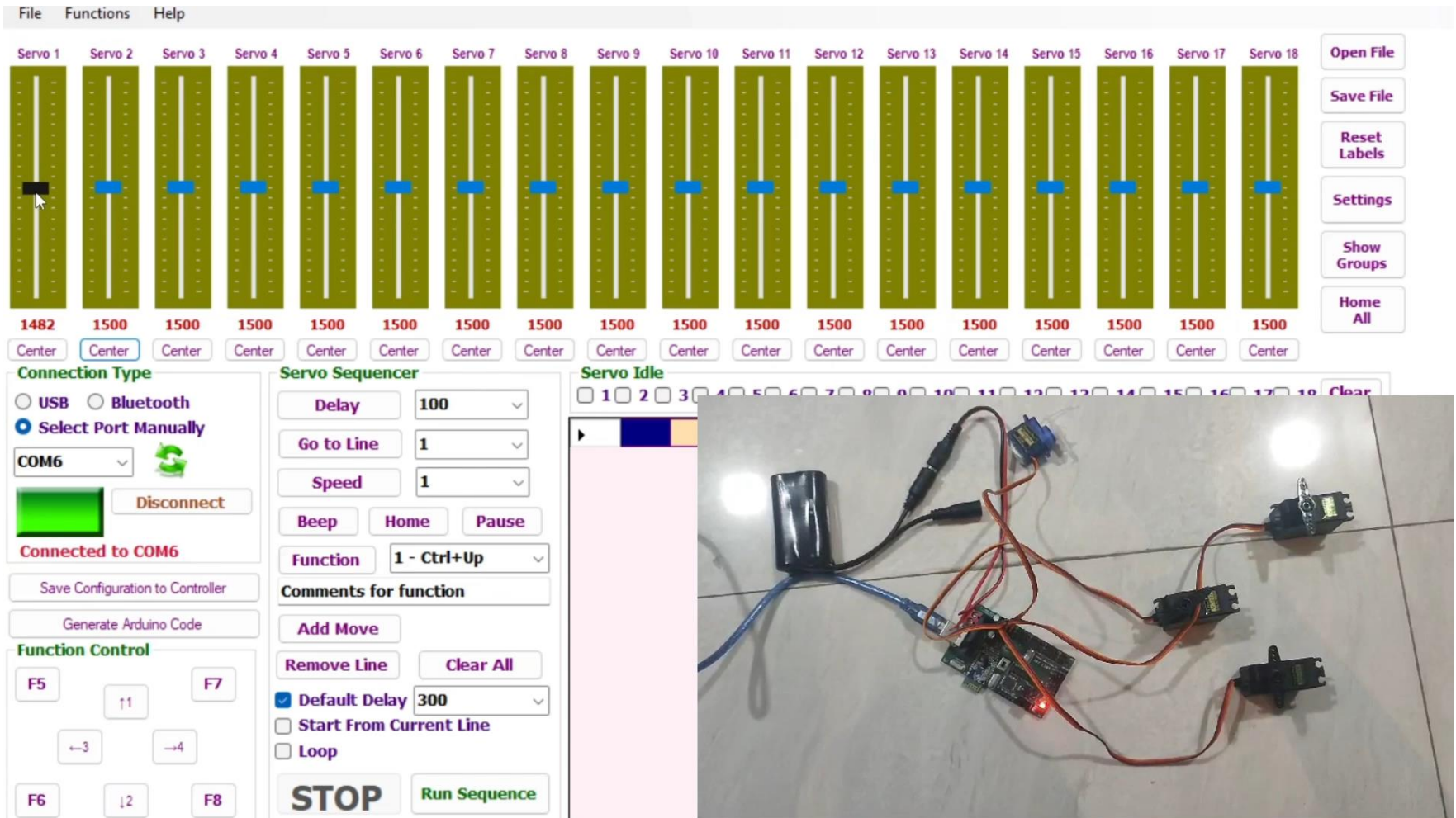
- **All components have been connected.**



Progress – System Testing

- Motor Setup:** Successfully connected and tested three MG995 servo motors along with the SG90 micro servo.
- Control Method:** Using the servo controller software, we ran tests to control the motors.

Demonstration Video



Calculations

1. Torque Calculation For Servo Motors

$$T = F \times r$$

where,

T = Torque (Nm)

F = Force applied (N) — 4.90 N

r = Distance from the axis of rotation (cm) — 22cm

$$T = 4.90 \text{ N} \times 22 \text{ cm}$$

$$T = 108 \text{ N-cm}$$

2. Pay load Capacity

$$\text{Max load} = \frac{\text{Torque}}{\text{Distance}}$$

$$= \frac{1.08 \text{ N-m}}{0.22 \text{ m}}$$

$$\text{Max load} = 4.90 \text{ N}$$

3. Power Consumption

$$P = V \times I$$

P = Power (W)

V = Voltage (V)

I = Current (A)

For an MG995 servo motor

- Operating Voltage = 6V

- Current (No load) = 0.5 A

- Current (Under load) = 2.5 A

- We are using three MG995 servos :

$$P_{\text{total}} = 3 \times 6V \times 2.5 A$$

$$P_{\text{total}} = 45 W$$

- Battery Specification (Li-Ion)

Voltage = 7.4V

Current = 2600 mAh

4. Arm Reach and Angle Calculation

Considering

- Link length : 22 cm

- Angle : servo motors with (360°) angle rotation

$$\text{Total Reach} = \text{Link length} \times \cos(\theta)$$

$$\text{Total Reach} = 22 \text{ cm} \times \cos(30)$$

$$\boxed{\text{Total Reach} = 22 \text{ cm}}$$

5. Energy Consumption over time (V) $V = \text{Voltage}$
(A) $I = \text{Current}$

$$\text{Energy} = \text{Battery Capacity (Ah)} \times \text{Battery Voltage (V)}$$

Li-Ion battery

$$\text{Energy} = 2.6 \text{ Ah} \times 7.4 \text{ V}$$

$$\boxed{\text{Energy} = 19.24 \text{ Wh}}$$

IF servos consume 45 W total

$$\text{Battery life} = \frac{19.24 \text{ Wh}}{45 \text{ W}} = 0.4275 \approx 0.43$$

$$\boxed{(\text{Battery life} = 26 \text{ minutes})}$$

Next Steps

- **3D Printing:** Begin building and assembling 3D-printed parts for the robotic arm.
- **Modification:** Modify parts and assembly based on further testing and requirements.



Conclusion

- Our progress so far has built a solid base, making it easier to move forward with the next stages of development
- Moving forward with the building and testing phase to refine the arm's performance in cold storage.

THANK YOU